

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DODOMA REGION
CHAMWINO DISTRICT COUNCIL
FORM FOUR JOINT MID-TERM EXAMINATION
MAKANG'WA, MAKWAWA, MLOWA BARABARANI, MLOWA BWAWANI,
NHINHI, CHIBOLI, MANZASE AND FUFU SECONDARY SCHOOLS.**



032/1

CHEMISTRY 1

TIME: 2:30HRS

March, 2026:

INSTRUCTIONS

1. This paper consists of three sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in sections A and B and two question from section C.
3. Section A carries sixteen (16) marks, section B carries fifty-four (54) marks and section C carries thirty (30) marks.
4. Cellular phones are not allowed in the examination room.
5. Non-programmable calculators may be used.
6. Write your Examination Number on every page of your answer booklet(s).
7. The following constants may be used:

Atomic masses:

H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, Al = 27, S = 32, Cl = 35.5,
K = 39, Cu = 64

Avogadro's number = 6.02×10^{23} . GMV at S.T.P. = 22.4 dm^3 .

1 Faraday = 96,500 coulombs.

Standard pressure = 760 mm Hg, Standard temperature = 273 K.

1 Litre = $1 \text{ dm}^3 = 1000 \text{ cm}^3$.

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SECTION A: (16 marks)

Answer all questions in this section

1. For each of the item (i) to (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the booklet provided.
 - i. Which fuels among the following are originated from fossil fuels?
A. Firewood and coal B. Coal and firewood C. Petrol and coal
D. Petrol and charcoal E. Diesel and firewood
 - ii. The granules of solid substances can be grinded into fine powder in the laboratory by:
A. Pestle and filter funnel B. Separating funnel and motor
C. Pestle and filter paper D. Thistle funnel and motor
E. Pestle and motor.
 - iii. An element X is found in period 4 and group II of the periodic table. If element X undergoes the reaction $X \longrightarrow X^{2+} + 2e^-$. The electronic configuration of X^{2+} ions is:
A. 2:8:6 B. 2:8:2 C. 2:8:8

2. Match the description in **List A** with the corresponding answers in **List B** by writing the letter of the correct response beside the item number in the booklet provided.

LIST A	LIST B
i. Biogas	A. Industrial gas which is a mixture of hydrogen and carbon monoxide
ii. Biomass	B. Non-renewable gaseous fuel
iii. Natural gas	C. Industrial gaseous fuel which is a mixture of nitrogen gas and carbon monoxide
iv. Producer gas	D. Gaseous fuel derived from decomposition of biological wastes
v. Water gas	E. A fuel derived from plants such as cassava, sorghum and sugar cane
vi. Ethanol	F. Renewable sources of energy
	G. Water gas and ammonia
	H. Ammonia and butane

SECTION B: (54 marks)

Answer all questions in this section

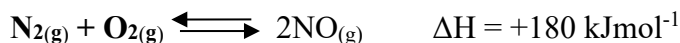
3. (a) Write balanced equation of
- Sodium hydroxide react with Sulphuric acid
 - Calcium carbonate decomposed by heat
- (b) (i) With aid of a balanced chemical equation name the products formed when nitrates of potassium and zinc decompose by heat.
- (ii) Suggest why nitrates of zinc and potassium behave differently on heating.

4. (a) The following table contain some names of laboratory apparatus

Tripod stand	Combustion tube
Measuring cylinder	Retort stand and clamp
Pipette	Wire gauze
Petri dish	Burette
Conical flask	Bunsen burner

Using the names given in the table above, identify five apparatus that are used in volumetric analysis and suggest their functions.

- (b) 6.9g of Y_2CO_3 was dissolved in water to make 0.5 litres of a solution. 25cm^3 of this solution required 50.0cm^3 of 0.1M HCL for complete neutralization.
- Write the balanced chemical equation for the reaction
 - Calculate the concentration of Y_2CO_3 in mol/dm^3
 - Calculate the relative molecular mass of Y_2CO_3
 - Calculate the relative atomic mass of Y
 - What is the name and symbol of element Y
5. (a) State five characteristics of a homologous series
- (b) Draw the open structure formula of 2,2-dichlorohexane
- (c) Giving two reasons, explain why 2, 2-dichloro-3-methyl butane is a structural isomer of 2,2-dichloropentane.
6. A colourless gas G fumes in the air. The gas gives dense white fumes when a glass rod dipped in ammonia solution is held near the glass.
- Name the gas G.
 - Name two chemicals used in the preparation of gas G.
 - Write a balanced chemical equation for the reaction of the chemicals named in b.
 - Why does the gas G fume strongly in air?
 - Why does the gas G form dense white fumes with ammonium hydroxide?
7. (a) Consider the formation of nitrogen monoxide using the reversible reaction.



In order to obtain maximum yield of nitrogen monoxide at equilibrium it is necessary to control the conditions very carefully. State with reason;

- (i) Whether a high or low temperature should be used.
 - (ii) If any increase in pressure would have effect on the rate of the reaction.
 - (iii) Whether the addition of nitrogen gas from outside the system will have any effect to the production of nitrogen monoxide.
- (b) Name three factors which will determine the production of substances released in the electrodes during electrolysis.
8. (a) Explain the following
- (i) Why group I elements are called alkali metals
 - (ii) Why non-metals are called oxidizing agents
- (b) The reaction between dilute hydrochloric acid and sodium thiosulphate results in the production of white precipitates which cause the opaqueness of the solution. Mention any three (3) uses of solid substance which cause the opaqueness of the solution.

SECTION C: (30 MARKS)

Answer only two questions from this section

9. Chibago had a visit in Mgunda's maize farm, he found Mgunda using much manure over fertilizer on his cultivation, and Chibago advised him to use fertilizers over manure. As a young scientist oppose Chibago's argument using five points.
10. When dilute nitric acid is added to a given solid P, blue solution X is formed and a gas R that forms a white precipitate with lime water. Heating blue solution of the hydrated salt X, it initially loses its water of crystallization and then decomposes to form a black solid S, brown fumes of gas Y and gas Z that relight a glowing splint.
 - a) Identify the solid X and gases Y and Z
 - b) Write the balanced chemical equation for the reaction between the solid P and dilute nitric acid
 - c) Give a balanced chemical equation for the formation of the solid S.
 - d) Describe two chemical properties of gas Z
 - e) Give the IUPAC name of a green solid P
11. Using three points in each briefly explain the effects and ways of preventing land pollution.